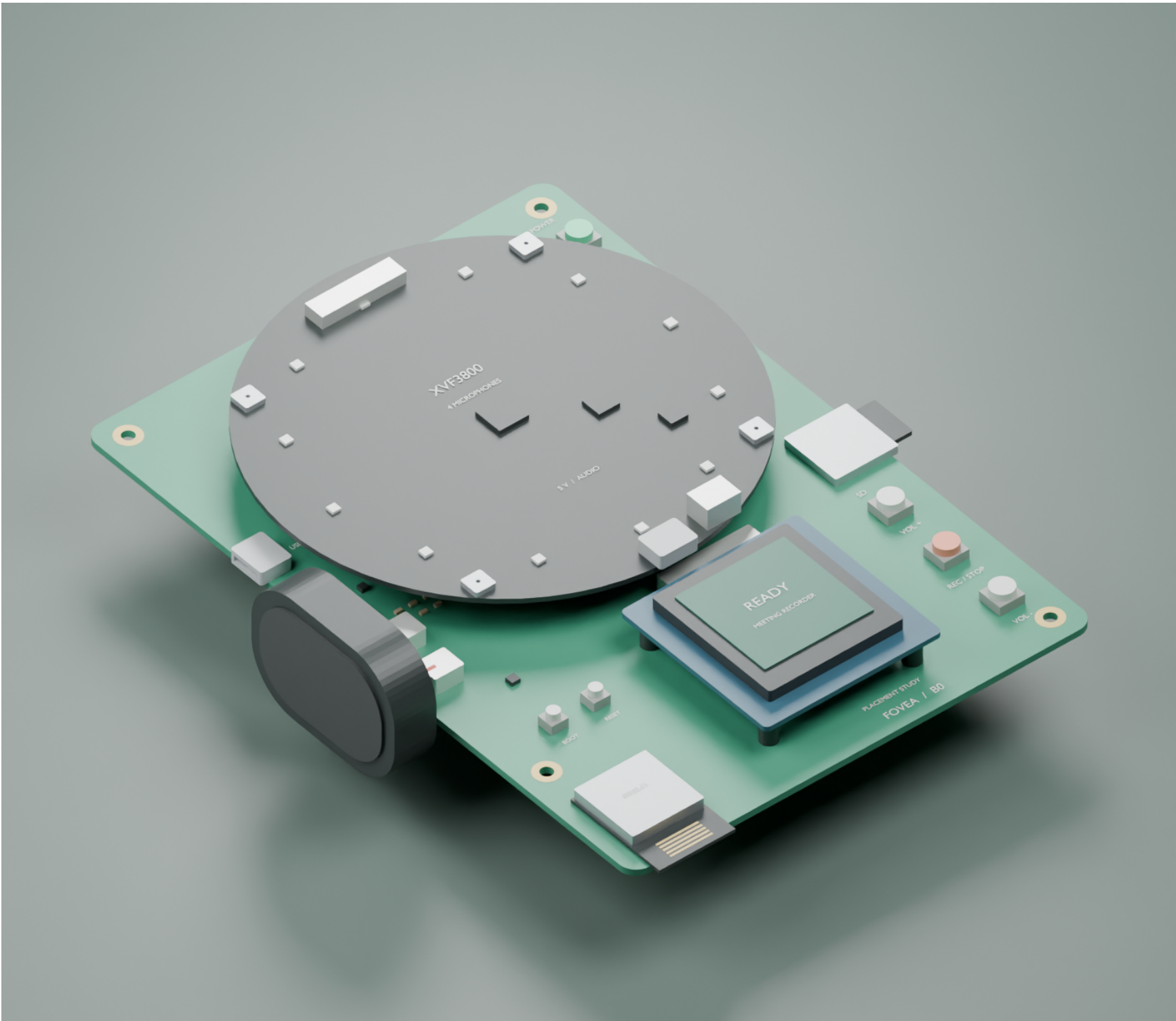


Custom board, microphone array and controls

A dimensioned placement model plus an editable circuit draft. The microphone module is the agreed first-prototype route.



MAIN PCB

110 x 158 x 1.6 mm. Proposed four-layer board; copper is not routed.

AUDIO

100 mm ReSpeaker XVF3800 module. Four installed microphones in a 66 mm square. One side-facing speaker.

SCREEN AND BATTERY

1.3-inch SPI module envelope; measured 91 x 59 x 5 mm battery below the main board.

SIX BUTTONS

Power, Record/Stop, Volume +, Volume -, BOOT and RESET. Switch actuators and case caps need final part selection.

MODEL ACCURACY

Board outline is dimensioned. Component bodies, supports and wiring are clearance models; connector mates and screen holes are not released.

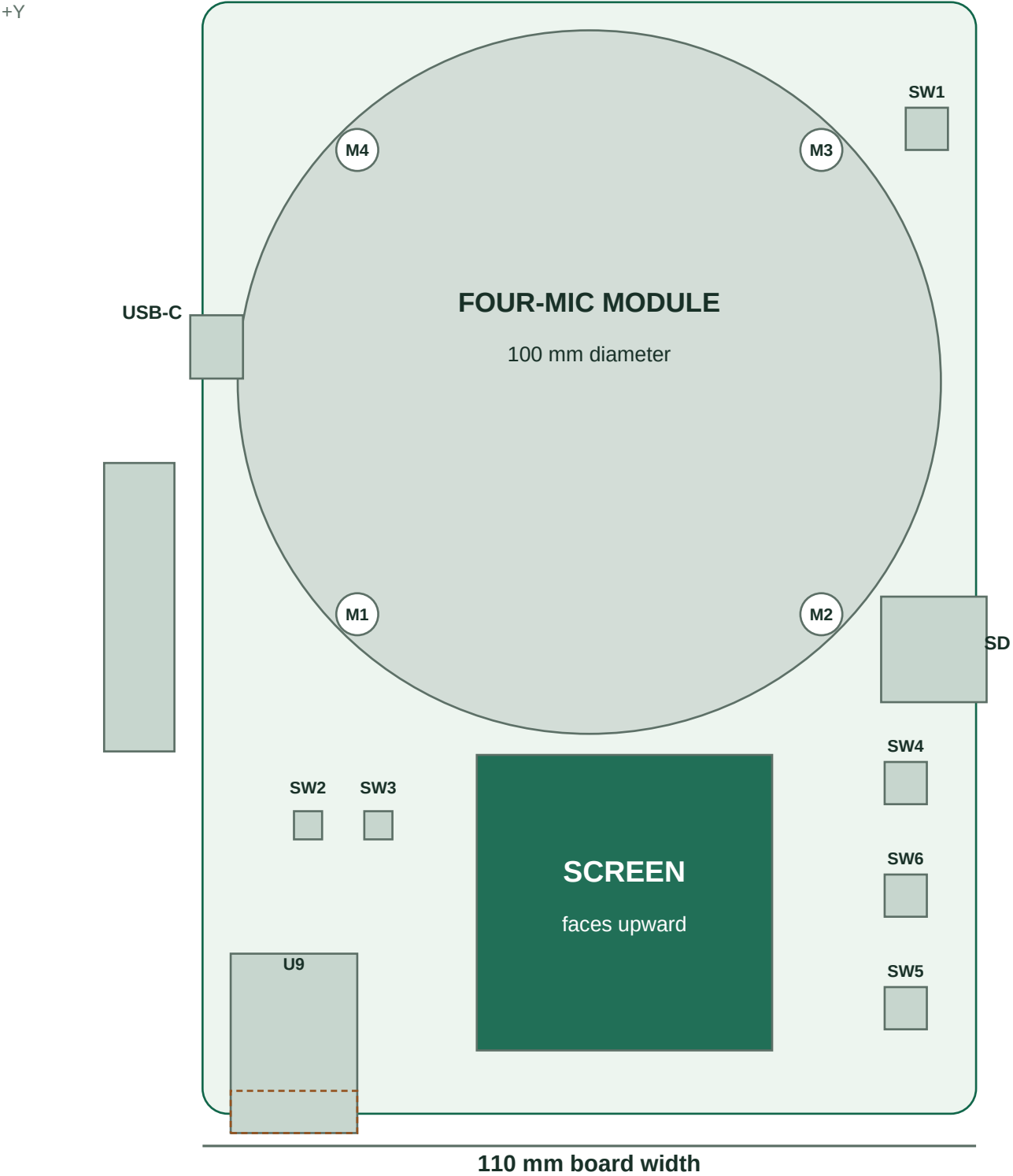
MANUFACTURING STATUS

No Gerbers, routed PCB, ERC pass or physical validation. Use this package for engineering review, not a PCB order.

Board map: what faces upward

View from above, with the recorder lying flat on a table. References match the circuit sheets later in this PDF.

TOP END



XY drawing follows the current 3D model. Not a drill or copper plot.

MICROPHONES M1-M4

The four small microphones sit on the round XVF3800 board. Their acoustic openings and the screen face the same upward direction. The round board is not a speaker.

USB-C J1: LEFT EDGE

The existing model puts the charging/programming connector on the left edge. A cable enters horizontally. Keep the opening and plug clearance clear of the microphone grille.

CARD J4: RIGHT EDGE

The microSD card slides out through the right side. Leave space for the card, fingertip and push-push travel in the enclosure.

SPEAKER: LEFT SIDE

The measured oval speaker faces outward through a side grille, below the USB-C area. Neither the table nor the back cover should cover its sound opening.

BATTERY: UNDER THE MAIN BOARD

The 91 x 59 x 5 mm pouch is beneath insulating support. Allow retention, wire clearance and pack tolerance without compressing the pouch.

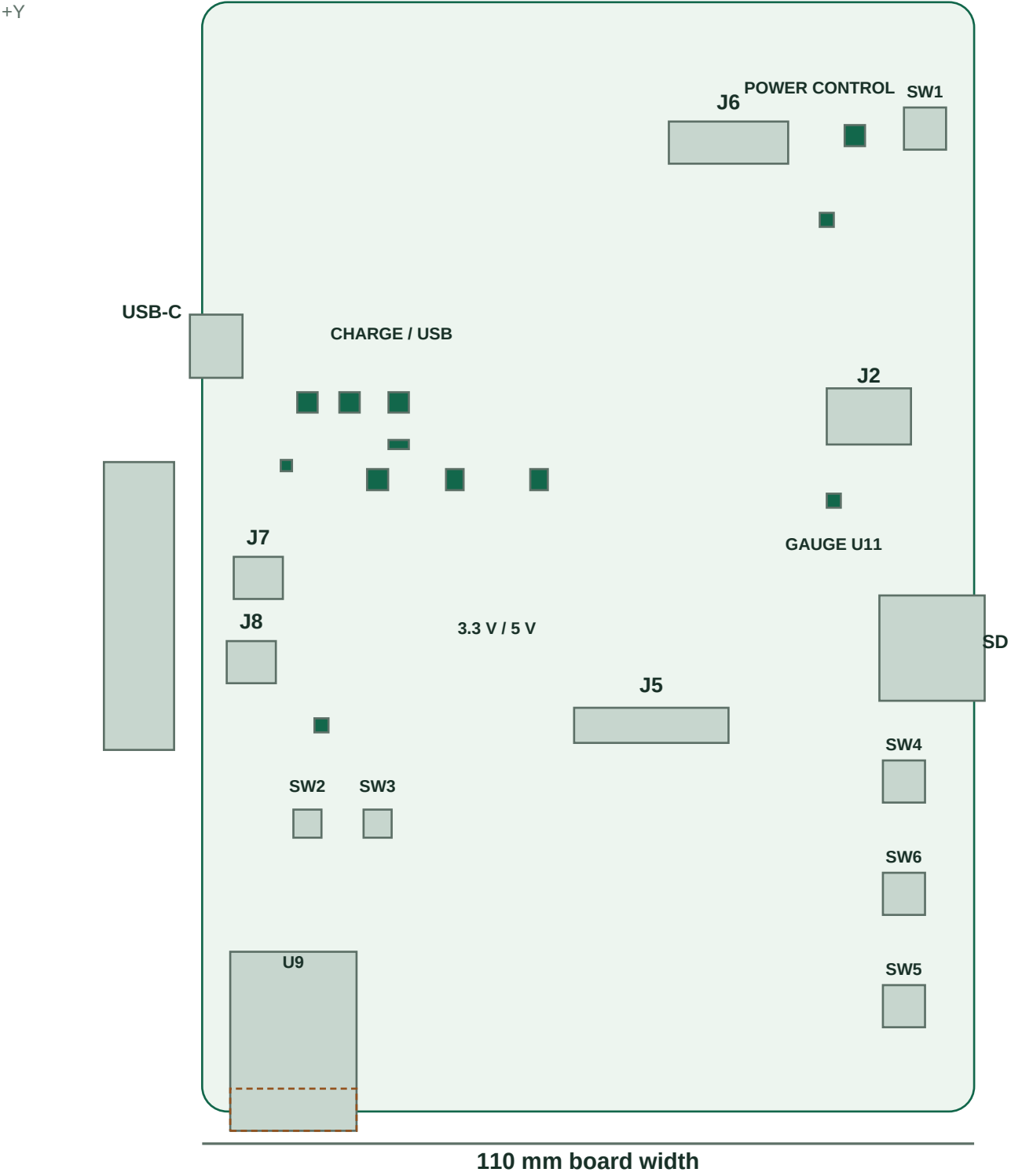
DO NOT MOVE THE FOUR SENSORS INDIVIDUALLY

This purchased module fixes the microphone spacing. Keep the screen and solid case ribs away from the four acoustic openings. Loose microphones would require a different design.

Under the array: main-board component locations

Audio module and display removed in this view. Positions are nominal envelopes, not approved footprints.

TOP END



XY drawing follows the current 3D model. Not a drill or copper plot.

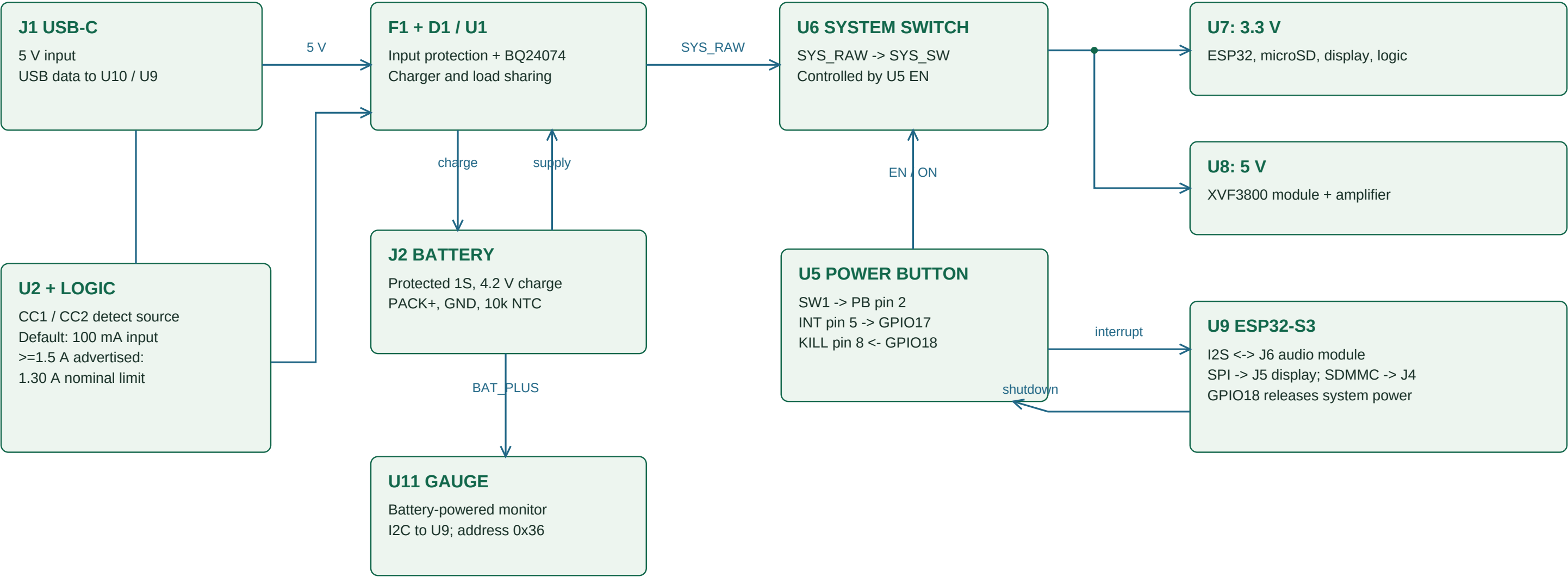
Ref	What / where	Centre X,Y (mm)
J1	USB-C, left edge	-53, +30
J2	Battery + NTC, upper right	+40, +20
J3	UART service pads	Unplaced
J4	microSD socket, right edge	+49, -13
J5	Display harness, below array	+9, -24
J6	Audio logic harness, upper right	+20, +59
J7 / J8	Amplifier in / speaker out, left	-47,-3 / -48,-15
U1 / U2	Charger / USB-C current detector	-30,+11 / -43,+13
U5 / U6	Power controller / load switch	+38,+60 / +34,+48
U7 / U8	3.3 V / 5 V converters	-19,+11 / -7,+11
U9	ESP32-S3 module, lower left	-42, -69
U10 / U11	USB isolation / battery gauge	-38,-24 / +35,+8
U3 / U4 / U12 / Q1	USB suspend/current logic	See model envelopes
SW1	Power, upper right	+48, +61
SW2 / SW3	BOOT / RESET, lower left	-40,-38 / -30,-38
SW4 / SW6 / SW5	Volume + / Record / Volume -	+45,-32 / -48 / -64

Origin: main-board centre. +X is right; +Y is toward the microphones. All values describe the current CAD placement study.

U9 is rotated 180 degrees relative to the pin map on the following pages: its antenna faces the bottom edge. Final antenna keepout, connector key orientation, passives and mounting holes must be checked after footprint selection.

Power distribution and system interconnect

Arrows show power or signal flow. All grounds join the board ground system; keep switching and speaker currents away from the microphone interface.



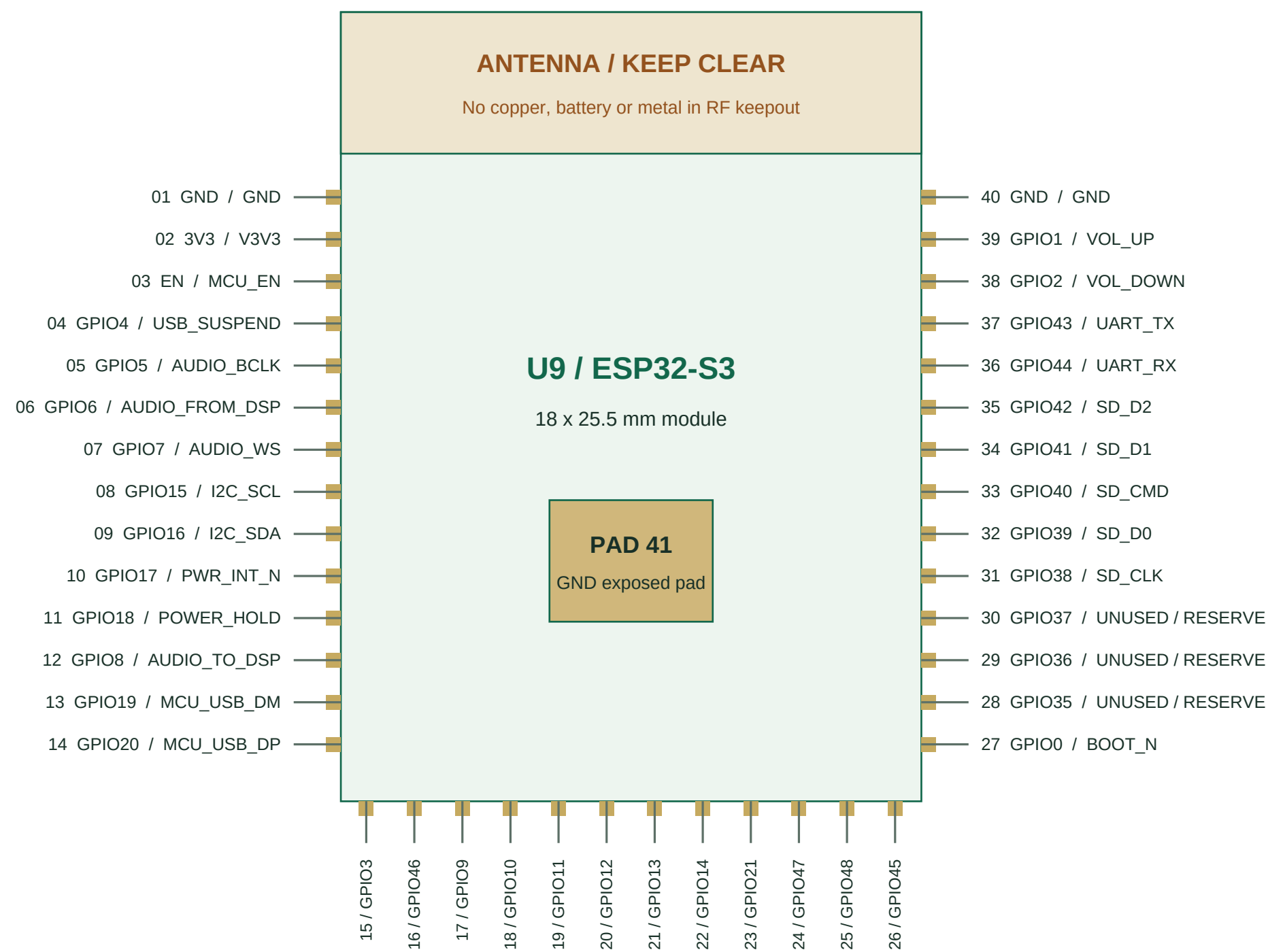
RAIL BUDGETS

Rail	Recording / peak	Purpose	Important limit
V3V3	0.33 A / 0.85 A	Digital logic and storage	3.3 V logic; do not connect to 5 V
V5_AUDIO	0.30 A / 0.871 A	Mic DSP and speaker amplifier	Budget assumes <=2 W speaker output
BAT_PLUS	0.80 A / 3.07 A estimated	Pack discharge incl. path losses	>=4 A pack/harness target; verify rating
SYS_RAW	Battery-dependent / ~4.4 V on USB	Before main power switch	Charger and control remain on when off

Electrical details and all bypass components are defined in the complete net sheets. Current-source detection is not USB-PD negotiation. The USB socket is not a USB microphone host port in this revision.

ESP32 module: physical pad map

U9 ESP32-S3-WROOM-1-N16R8. Top view of the module, antenna upward. The module is rotated 180 degrees in the board placement.



Pad number is the physical solder pad; GPIO is the programmable signal name. Example: module pad 8 is GPIO15 (I2C SCL), not GPIO8. Check the table before wiring.

Source: Espressif module datasheet, Pin Layout and Pin Definitions. GPIO35-37 are reserved for this N16R8 memory configuration. GPIO0 is the BOOT strap. This drawing is not a footprint.

ESP32 pad-to-function wiring schedule

Physical module pad, GPIO name, assigned net and destination. These are custom-board assignments; existing touchscreen firmware uses a different profile.

Pad	GPIO / pin	Net name	Connects to
1	GND	GND	Ground plane
2	3V3	V3V3	U7 3.3 V / local bypass
3	EN	MCU_EN	SW3.1, J3.4
4	GPIO4	USB_SUSPEND	Q1.1
5	GPIO5	AUDIO_BCLK	J6.3
6	GPIO6	AUDIO_FROM_DSP	J6.7
7	GPIO7	AUDIO_WS	J6.5
8	GPIO15	I2C_SCL	J6.10, U11.7
9	GPIO16	I2C_SDA	J6.9, U11.8
10	GPIO17	PWR_INT_N	U5.5
11	GPIO18	POWER_HOLD	U5.8
12	GPIO8	AUDIO_TO_DSP	J6.8
13	GPIO19	MCU_USB_DM	R33 -> U10.2
14	GPIO20	MCU_USB_DP	R32 -> U10.1
15	GPIO3	NC	Leave unused / reserve
16	GPIO46	NC	Leave unused / reserve
17	GPIO9	AUDIO_RESET_N	J6.11
18	GPIO10	LCD_CS	J5.5
19	GPIO11	LCD_MOSI	J5.3
20	GPIO12	LCD_CLK	J5.4
21	GPIO13	LCD_DC	J5.6

Pad	GPIO / pin	Net name	Connects to
22	GPIO14	LCD_RST	J5.7
23	GPIO21	RECORD_N	R55 -> SW6; R56 / C64
24	GPIO47	LCD_BL	J5.8
25	GPIO48	SD_D3	J4.2
26	GPIO45	NC	Leave unused / reserve
27	GPIO0	BOOT_N	SW2.1, J3.5
28	GPIO35	NC	Leave unused / reserve
29	GPIO36	NC	Leave unused / reserve
30	GPIO37	NC	Leave unused / reserve
31	GPIO38	SD_CLK	R40 -> J4.5
32	GPIO39	SD_D0	J4.7
33	GPIO40	SD_CMD	J4.3
34	GPIO41	SD_D1	J4.8
35	GPIO42	SD_D2	J4.1
36	GPIO44	UART_RX	J3.3
37	GPIO43	UART_TX	J3.2
38	GPIO2	VOL_DOWN	R53 -> SW5; R54 / C63
39	GPIO1	VOL_UP	R51 -> SW4; R52 / C62
40	GND	GND	Ground plane
41	EP	GND	Ground plane

USB and SD clocks have series resistors: connect to the named MCU-side net, not directly across the resistor. GPIO18 must behave as open drain for orderly power-off. GPIO4 controls USB suspend through Q1/U12.

Firmware must reserve GPIO35-37, preserve strap behaviour, choose the correct I2S clock role and map every button from this schedule. Signal direction is configured by firmware; a matching label alone does not establish it.

Connector pinouts: audio, screen and battery

All J2 / J5 / J6 assignments below are carrier-side definitions. Connector orientation, key and module-side mating pin numbers are NOT released.

J6 / FOUR-MIC MODULE HARNESS

Pin	Function	Net	Direction / level
1	+5V	V5_AUDIO	5 V power out
2	GND	GND	Ground
3	BCLK	AUDIO_BCLK	ESP -> DSP
4	GND	GND	Ground
5	WS	AUDIO_WS	ESP -> DSP
6	GND	GND	Ground
7	DSP OUT	AUDIO_FROM_DSP	DSP -> ESP
8	DSP IN	AUDIO_TO_DSP	ESP -> DSP
9	SDA	I2C_SDA	Bidirectional
10	SCL	I2C_SCL	ESP -> DSP / gauge
11	RESET_N	AUDIO_RESET_N	ESP -> DSP
12	GND	GND	Ground

BCLK, WS, data, reset and I2C use 3.3 V logic. Only J6 pin 1 is the 5 V supply. The array has its own four microphones; this is not four separate mic inputs.

Drawn as a keyed 2x6 logical interface. Do not use a guessed ribbon-cable pin order: the exact connector and Seeed module contact mapping must be checked together.

PIN-1 CONVENTION

A final assembly drawing must show each connector from a named viewing direction and identify pin 1 with a triangle or square pad. These tables do not claim an existing plug is wired in this order.

J5 / SPI SCREEN MODULE

Pin	Function	Net	Controller / supply
1	VCC	V3V3	U7 3.3 V / local bypass
2	GND	GND	Ground plane
3	DIN	LCD_MOSI	U9.19
4	CLK	LCD_CLK	U9.20
5	CS	LCD_CS	U9.18
6	DC	LCD_DC	U9.21
7	RST	LCD_RST	U9.22
8	BL input	LCD_BL	U9.24

Requires a 3.3 V-compatible module with a backlight driver. BL is a control signal, not a direct high-current LED feed.

J2 / PROTECTED BATTERY + TEMPERATURE

Pin	Function	Net	Rating / note
1	PACK+	BAT_PLUS	Protected 1S; 4.2 V charge
2	PACK-	GND	Ground return; >=4 A path target
3	NTC	BAT_NTC	Compatible 10k thermistor to PACK-

Battery capacity does not prove discharge current. The old two-wire pack is not electrically approved for this three-wire charging/NTC arrangement.

Connector pinouts: USB, SD, service and speaker

USB names identify standard receptacle contacts; SD names identify card contacts. Translate these to the exact socket footprint before routing.

J1 / USB-C USB 2.0

Contacts	Net	Connection
A4 A9 B4 B9	VBUS_5V	5 V input -> F1 -> USB_PROTECTED
A1 A12 B1 B12	GND	Ground
A5	CC1	U2.1 + D3.1
B5	CC2	U2.2 + D3.2
A6 B6	USB_DP	U10.8 + D4.1
A7 B7	USB_DM	U10.7 + D4.2
A8 B8	NC	SBU unused

Shield / S1 GND Exact shell pad count depends on socket

J3 / SERVICE PADS

Pad	Net	Use
1	GND	Debugger ground
2	UART_TX	U9 pad 37 / GPIO43 -> adapter RX
3	UART_RX	U9 pad 36 / GPIO44 <- adapter TX
4	MCU_EN	Reset / enable
5	BOOT_N	GPIO0 boot strap
6	V3V3	Voltage reference only; do not supply power

Use 3.3 V UART levels. A USB-UART adapter must not power the switched-off board through its TX line or VREF. Service-pad location is still unplaced.

J4 / microSD CARD CONTACTS

Pin	Card	Net	ESP32 destination
1	DAT2	SD_D2	U9.35
2	DAT3	SD_D3	U9.25
3	CMD	SD_CMD	U9.33
4	VDD	V3V3	U7 3.3 V / local bypass
5	CLK	SD_CLK_CARD	R40 -> GPIO38 / U9.31
6	VSS	GND	Ground plane
7	DAT0	SD_D0	U9.32
8	DAT1	SD_D1	U9.34
S	SHELL	GND	Ground plane

J7 / AMPLIFIER IN -> J8 / SPEAKER OUT

Module-side input	Speaker output	Signal
J7.1	J8.1	SPK_PLUS
J7.2	J8.2	SPK_MINUS

The module amplifier drives both speaker wires. Neither speaker terminal is ground. Target a documented 4-ohm speaker and limit output to the assumed 2 W until thermal and current tests pass.

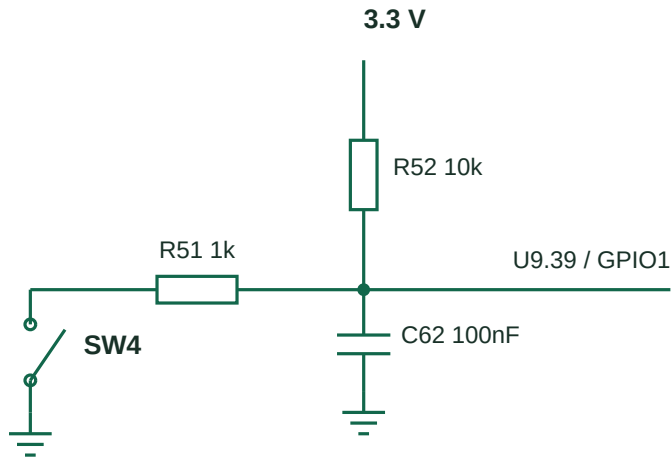
Card-detect and multiple shield pads depend on the selected SD socket. They are not interchangeable with the eight card contacts. The speaker harness must match the actual module connector.

Button circuits and safe shutdown

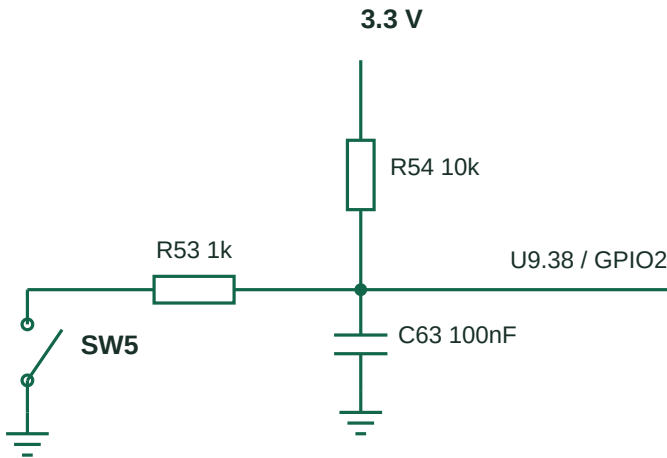
Momentary switches connect to logic inputs. The user power button does not carry the battery load current.

VOLUME AND RECORD / STOP: THREE IDENTICAL INPUT CIRCUITS

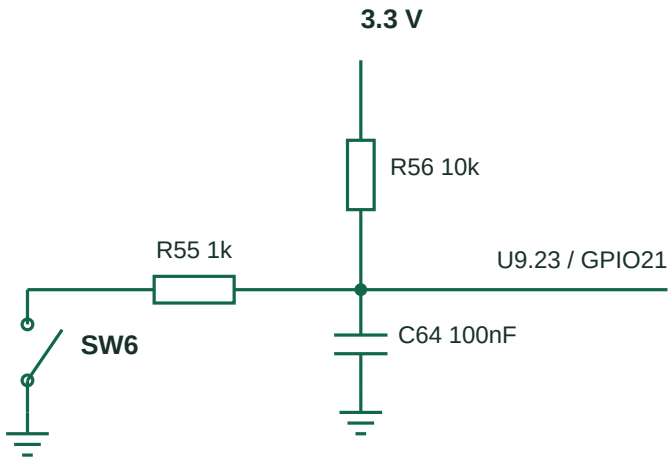
SW4: VOL_UP



SW5: VOL_DOWN

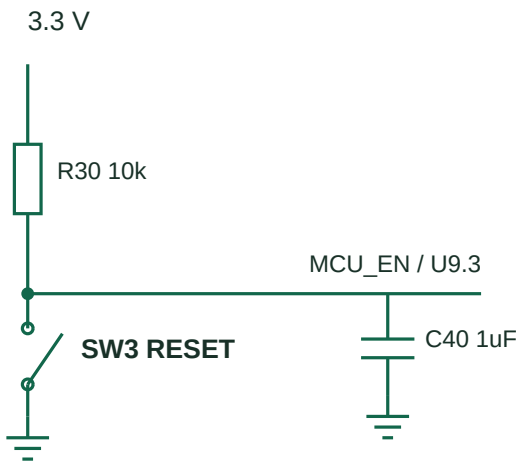
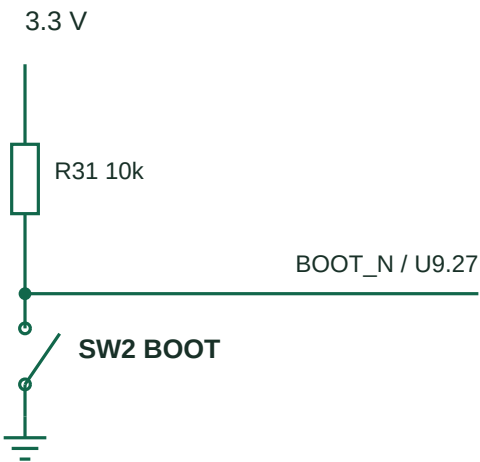


SW6: RECORD_N



Press = LOW. Current while held is about 0.30 mA; pin voltage is about 0.30 V. Firmware must debounce the input. Use the same net labels shown in the pad schedule.

BOOT / RESET



SW1 POWER / LTC2954 U5

SW1 shorts U5 pin 2 PB to GND; use the internal button input circuitry.
U5 pin 5 INT -> U9 pad 10 / GPIO17. R11 pulls INT to 3.3 V.
U5 pin 8 KILL <- U9 pad 11 / GPIO18, open drain. R12 pulls it to switched 3.3 V.
U5 pin 6 EN -> U6 pin 3 ON; R10 pulls EN to SYS_RAW.

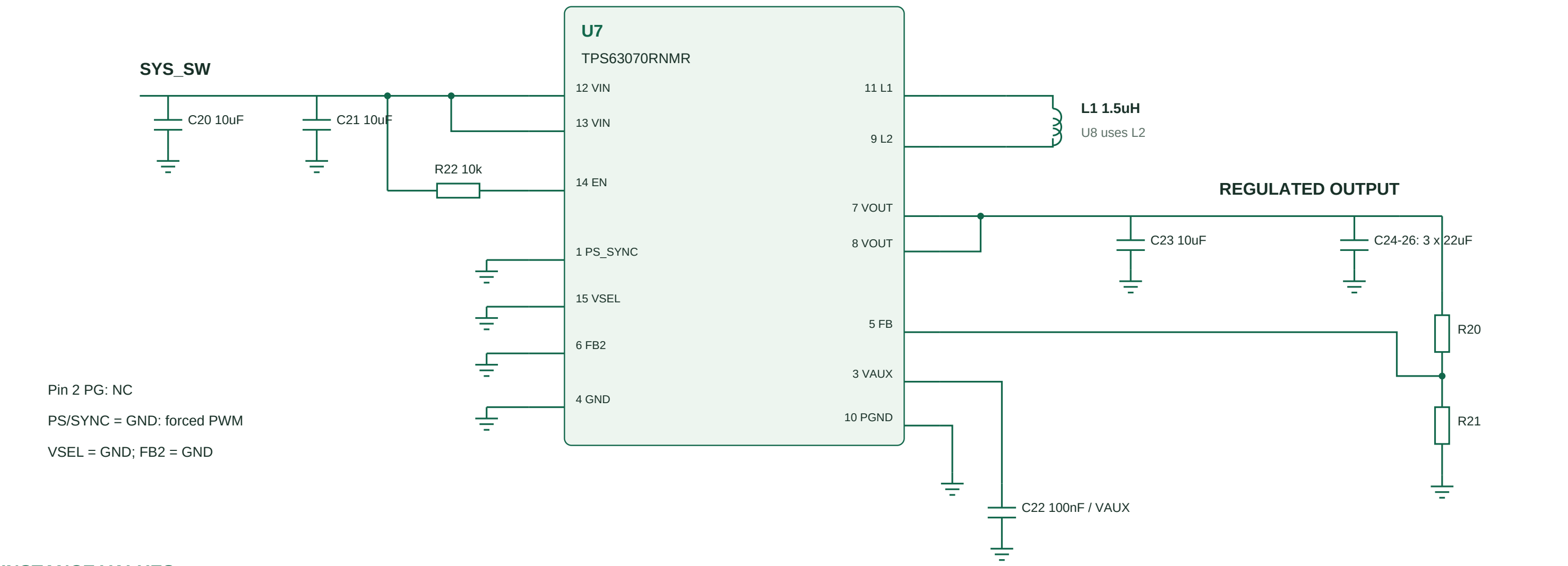
C11 47nF sets approximately 0.33 s start hold; C12 1uF sets approximately 6.48 s forced-off hold. Timings include the datasheet fixed delays and remain subject to tolerance.

SHUTDOWN ORDER

Power press -> finish recording -> finalize and flush WAV -> persist upload state -> mute -> GPIO18 LOW -> main rails off.
Forced off can interrupt writes. Charging/control/gauge stay powered; this is not a total battery disconnect.

Regulator wiring: two copies of this circuit

Drawn once for clarity; the table gives U7 and U8 component references. Every pin is also listed in the complete net sheets.



INSTANCE VALUES

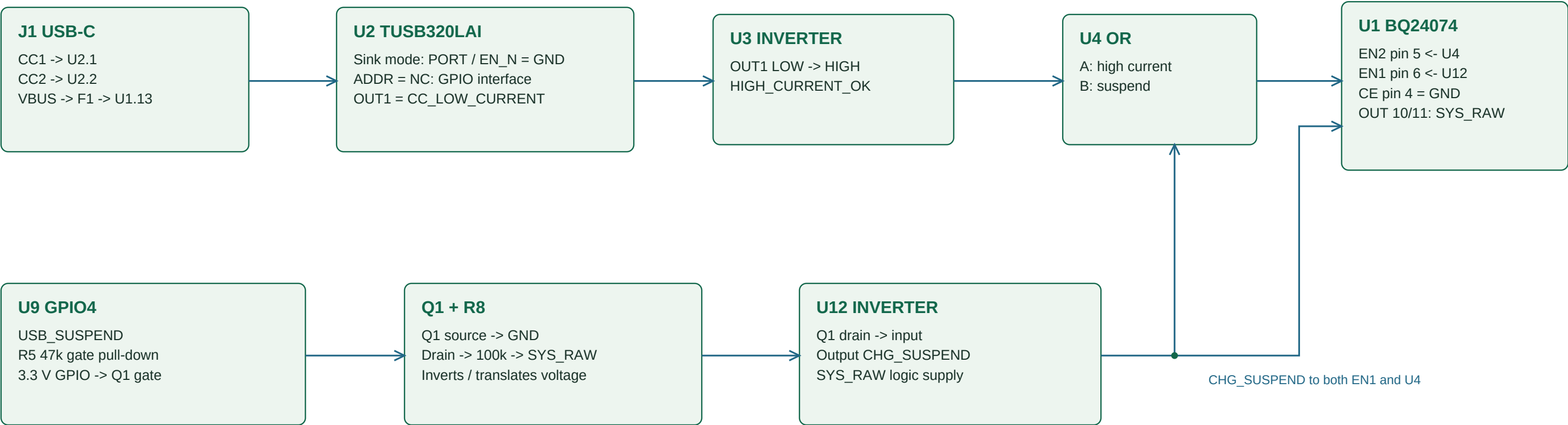
IC	Rail	Inductor	Feedback top / bottom	Enable resistor	Input / VAUX caps	Output capacitors
U7	3.3 V	L1 1.5uH	R20 470k / R21 150k	R22 10k	C20 C21 / C22	C23 + C24 C25 C26
U8	5 V	L2 1.5uH	R24 680k / R25 130k	R26 10k	C30 C31 / C32	C33 + C34 C35 C36

Feedback resistors: 0.1%. Use the selected part datasheet layout, short switching loops and appropriate effective ceramic capacitance. U7/U8 are 15-pad VQFN-HR packages, not generic QFN16.

OPEN: computed peak-load right-half-plane zeros are below the vendor guideline. The values shown are draft calculations; loop stability and load-step performance must be resolved before manufacture.

USB current selection and battery charging

Connected functional schematic. The complete component sheets include every physical pin, bypass capacitor, protection device and unused pin.



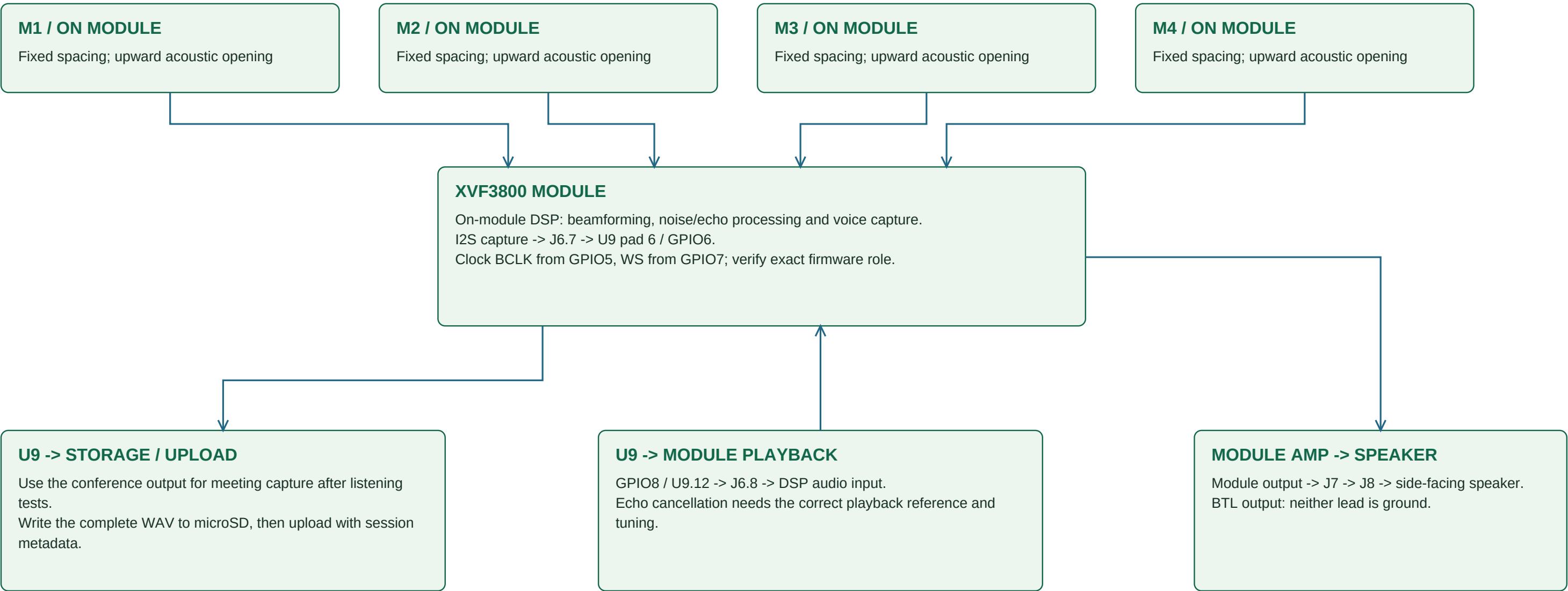
Source mode	GPIO4 suspend	EN1	EN2	BQ24074 input policy
Default / absent	0	0	0	USB100: 100 mA mode
Advertised >=1.5 A	0	0	1	ILIM resistor: 1.30 A nominal (1.40 A upper estimate)
Any	1	1	1	USB input suspended; battery may still supply SYS_RAW

U1 battery pins 2/3 -> J2.1; ground -> J2.2; TS pin 1 -> J2.3 compatible 10k NTC to pack ground. R2 1.78k from ISET pin 16 to ground sets about 0.50 A charging. R1 1.24k from ILIM pin 12 to ground sets the advertised-source input limit.

TUSB320LAI supplies CC sink termination: do not add parallel external Rd resistors. R3 is 900k from VBUS to U2 VBUS_DET. USB data follows J1 -> U10 isolation -> R32/R33 -> U9 GPIO20/19. No USB microphone-host support is implemented by this circuit.

Audio signal path and microphone expectations

Four physical microphones feed the purchased DSP module. Its processed channels are not four automatically identified people.



EXPECTED QUALITY AND ACCEPTANCE TEST

The manufacturer advertises 360-degree capture up to 5 m and onboard acoustic processing. That supports testing this module for meetings, but is not proof of clear speech in your enclosure, a noisy cafe or a reverberant room. A 3-5 m capture distance remains a target.

Before a custom PCB order: record the same passage at 1 m, 3 m and 5 m; rotate the source around the case; include fan noise and two people alternating or overlapping. Listen to the saved WAV and compare transcript omissions, clipping and speaker separation against the existing board.

Standard I2S firmware documents channel 0 as Conference and channel 1 as ASR. Other firmware images route channels differently. Source: Seeed XVF3800 introduction / firmware guide (linked in references).

Prototype cost: budget before requesting quotes

MYR planning estimate for about five custom prototypes. Prices checked 22 September 2026; only the explicitly marked retail anchors are observed prices.

Item	Low RM	High RM	Basis / exclusions
Four-mic module	299	299	Observed Cytron listing, with enclosure; verify exact variant
ESP32-S3 module	25	45	Estimate; supplier and exact N16R8 part still to be quoted
Small screen	32	60	RM32 local display price anchor; final module not selected
Power, USB, SD socket, buttons, passives	120	240	Estimate from circuit complexity; excludes ESP32 above
Protected battery + NTC	45	90	Estimate; correct discharge rating / harness required
Speaker	10	25	Estimate; reuse existing only after rating check
microSD card	25	45	Estimate; selected reputable card
PCB fabrication, assembly, initial checks	250	600	Allocation per unit in a small batch; not a vendor quote
Printed case, cables, screws and supports	80	180	Estimate; case design and print method change cost

ESTIMATED BUILD COST: RM886-1,584 PER PROTOTYPE

Use roughly RM900-1,600 each for initial planning. This is not a procurement-ready BOM total. Shipping, tax, rejected parts, rework and extra PCB revisions are excluded.

ONE-TIME ENGINEERING ALLOWANCE

Provisional RM5,000-15,000 for schematic review, layout and prototype bring-up. This is our planning allowance, not a Malaysian supplier quote.
Major firmware work, acoustic tuning, certification and production tooling are additional.

FIVE-UNIT FIRST BUILD

Hardware allowance: approximately RM4,500-8,000.
With the engineering allowance: roughly RM9,500-23,000 before exclusions. Request itemized 2-unit and 5-unit quotations.
Test the RM299 audio module before committing to the custom board.

PRICE ANCHORS / MALAYSIAN QUOTATION ROUTES

Cytron: microphone module RM299

Cytron: 1.3-inch display example RM32

Nexus Avenue: PCB layout, prototype fabrication and assembly

LKL Sunrise: circuit design, layout and assembly in Malaysia

Voltage and current budget

Engineering allocations include conversion losses and simultaneous loads; actual selected hardware must be measured.

Load	Rail	Recording	Peak allocation	Basis
ESP32 + flash + PSRAM	3.3 V	200 mA	500 mA	Engineering allocation; Wi-Fi bursts included in peak
microSD	3.3 V	60 mA	250 mA	Card-dependent; measure selected card
SPI screen + backlight	3.3 V	60 mA	80 mA	Module provisional; brightness dependent
Other switched 3.3 V loads	3.3 V	10 mA	20 mA	Pull-ups/control and design allowance
XVF3800 module, silent	5.0 V	300 mA	400 mA	Allocation; no verified module current. RGB LEDs disabled
Speaker amplifier increment	5.0 V	0 mA	471 mA	2 W continuous test output / assumed 85% amplifier efficiency

Design result	Calculated value	Meaning / limit
3.3 V regulated rail	0.33 A recording / 0.85 A peak	Select and validate a >=1.0 A output stage across battery range.
5 V audio rail	0.30 A recording / 0.871 A peak	RGB LEDs off; speaker limited to the assumed 2 W test output.
Battery during recording	0.80 A / 2.97 W	At 3.7 V, 90% conversion efficiency, 0.15 ohm total path.
Battery at simultaneous peak	3.07 A / 9.84 W	At 3.2 V, 85% efficiency, 0.15 ohm path; >=4 A pack/harness target.
USB-C input limit	1.30 A nominal / 1.40 A upper estimate	Only after >=1.5 A Type-C advertisement; default mode is 100 mA.
Battery charging	0.50 A nominal; 0.443-0.553 A estimate	Protected 4.2 V cell and correct NTC mandatory; pack spec still open.
Illustrative runtime	4.0 hours	Assumes 4 Ah at 3.7 V and 80% usable energy; not a measured promise.
Charge duration	8 h ideal constant-current portion	CV taper, load sharing and temperature make full charging longer.

Peak speaker playback can exceed USB input power: battery supplement is required. This design does not promise maximum operation from an empty battery or a weak USB port. Do not infer a battery discharge rating from its mAh capacity.

Calculations and power-button behaviour

Values are derived from datasheets and explicit assumptions. They are not substitutes for load, thermal and stability tests.

Charge current

$I = 890 / 1780 = 0.500$ A. Including IC factor and 1% resistor extremes:
0.443-0.553 A.

USB current limit

$I = 1610 / 1240 = 1.298$ A. Worst factor / low resistor estimate = 1.401 A, below 1.5 A.

3.3 V feedback

$V = 0.8 \times (1 + 470k / 150k) = 3.307$ V. Static tolerance estimate
3.269-3.345 V.

5 V feedback

$V = 0.8 \times (1 + 680k / 130k) = 4.985$ V. Static tolerance estimate
4.926-5.043 V.

POWER BUTTON

Hold about 0.33 s to start. A normal press raises an interrupt: stop capture, finalize/flush WAV, persist upload state, mute, then drive POWER_HOLD low. Holding about 6.48 s forces off if firmware hangs. Timer tolerances apply. The charging/control circuits stay alive.

GPIO18 must be open-drain. The 100k switched-rail pull-up avoids depending on fast firmware boot to stay powered.

Battery path

Solve $I \times (V_{bat} - I \times R_{path}) = P_{load} / \text{efficiency}$. $R_{path} = 0.15$ ohm is an assumed total, not a measured value.

Speaker

At 2 W into 4 ohm: 2.83 Vrms, 0.707 Arms, 1.00 A sine peak in speaker wires. Neither wire is ground.

Buttons and bus

Volume/record 10k pull-up + 1k series: 0.30 mA held, input low 0.30 V.
I2C 4.7k: 0.70 mA low; ~398 ns rise at assumed 100 pF. Start at 100 kHz.

Recording storage

16 kHz x 16-bit mono = 32,000 bytes/s = 1.831 MiB/minute, plus the WAV header.

Checks that must close before fabrication

The calculated budget identifies concrete design risks. These remain release gates for the hardware engineer.

REGULATOR STABILITY

Estimated boost right-half-plane zeros at simultaneous peak: 283 kHz and 183 kHz. Both are below TI's 400 kHz guideline. Select effective output capacitance and prove loop/load-step stability; the shown 76 uF nominal bank is not a validated solution.

CHARGER TEMPERATURE

At 5 V USB, 0.5 A charging and the recording budget: approximately 1.04 W charger heat at a 3.7 V cell, or 1.39 W at 3.0 V. Using datasheet 44.5 C/W and 40 C ambient gives about 86 C / 102 C junction. Actual PCB/enclosure can be hotter; review charge-current reduction or a switching charger.

LOW BATTERY AND BURSTS

The peak case predicts only 2.74 V at converter inputs and 1.42 W in the assumed series path. Measure voltage droop and pack protection behaviour. Enforce battery/volume limits before accepting uninterrupted recording at this load.

UNKNOWN MODULE LOAD

The audio-module current is an allowance. Measure its startup, capture, playback and LED loads. If it exceeds the allocation, revise both converters, charging path and runtime estimate.

CONNECTOR AND FOOTPRINT RELEASE

USB suffix, battery pack/harness, SD shell/detect pads, display SKU, audio-module mating pin map, switch actuators and ESD/fuse selection remain open. Symbol net numbering alone does not define a physical connector.

POWER-OFF / FIRMWARE / RF

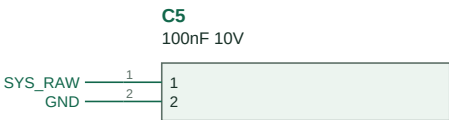
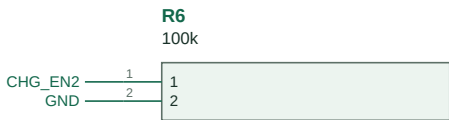
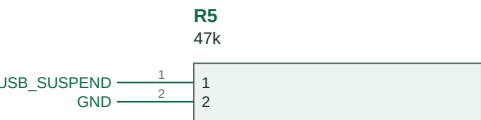
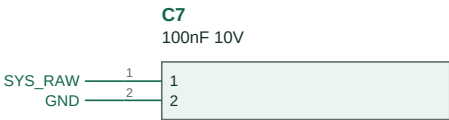
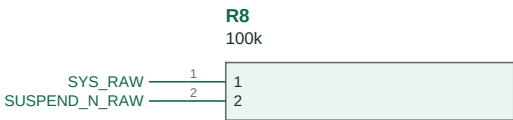
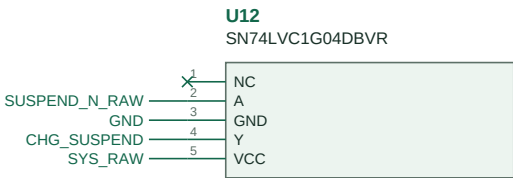
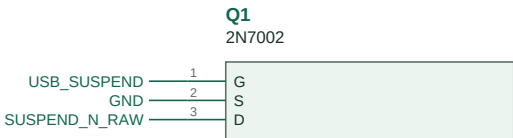
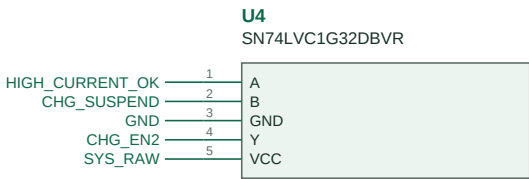
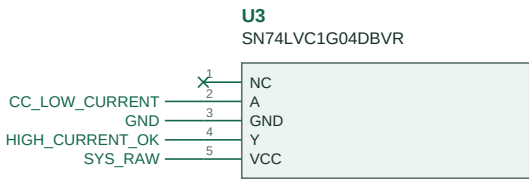
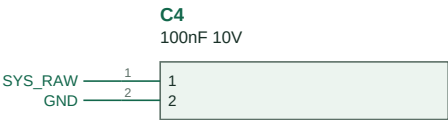
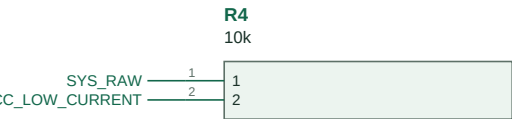
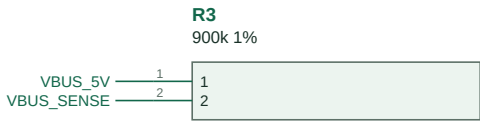
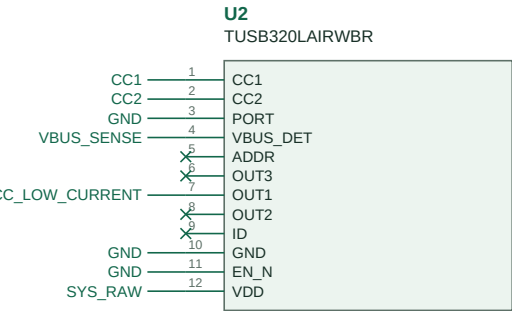
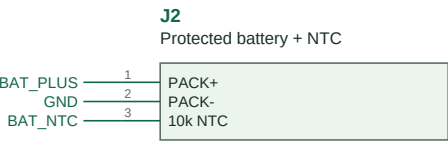
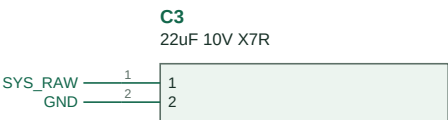
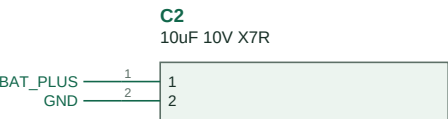
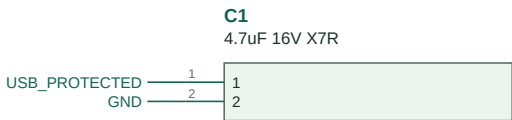
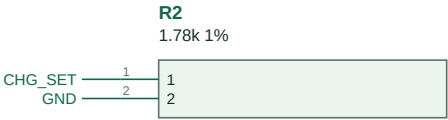
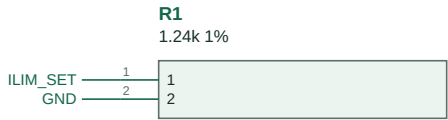
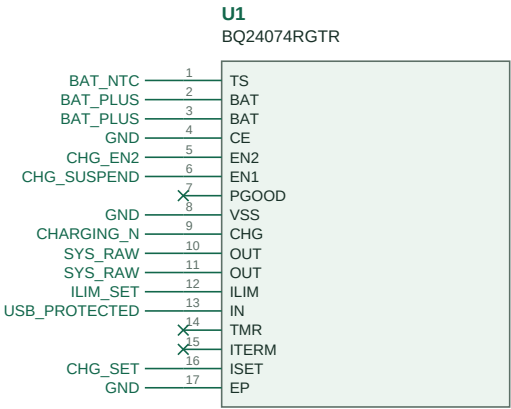
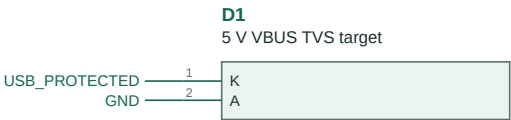
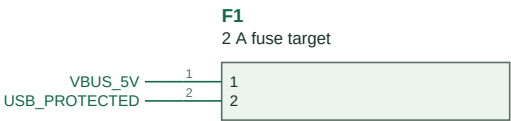
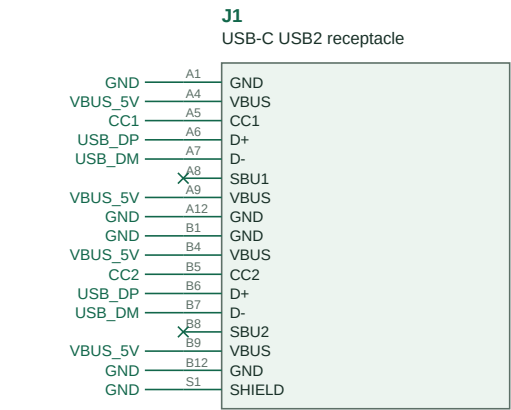
Validate no USB, gauge, debug or audio-module service-cable back-powering. Add the custom-board firmware profile and SD shutdown recovery. Keep battery, copper and metal away from the module antenna; verify the final RF keepout in CAD.

ENGINEERING VALIDATION

Run KiCad ERC with reviewed electrical pin types, assign verified footprints, route a four-layer board and run DRC. Test rails, battery temperature, USB source changes, SD integrity, recording range and echo cancellation. No native KiCad/ERC or hardware test has run here.

USB-C current detection and charging

Editable sheet: 01_usb_charge_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



TUSB320LAI supplies CC termination: do not add parallel 5.1k Rd resistors.

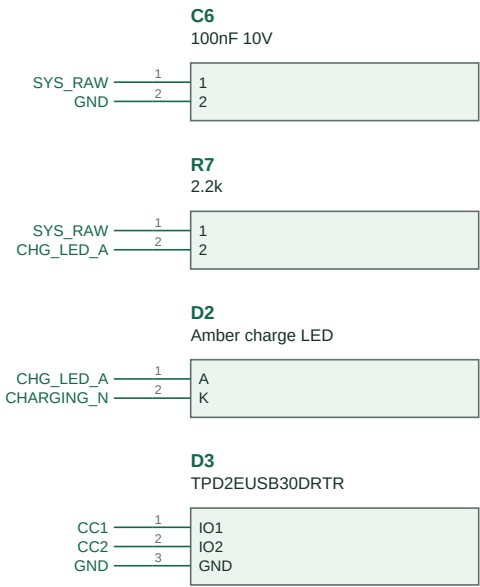
OUT1 low means 1.5 A or 3 A advertised. Inverter selects 1.30 A nominal input limit.

Default advertisement selects USB100. GPIO4 requests USB suspend; implement in firmware.

Battery: protected 1S, 4.2 V charge, >=4 A discharge target, 10k NTC. Existing pack not approved.

USB-C current detection and charging

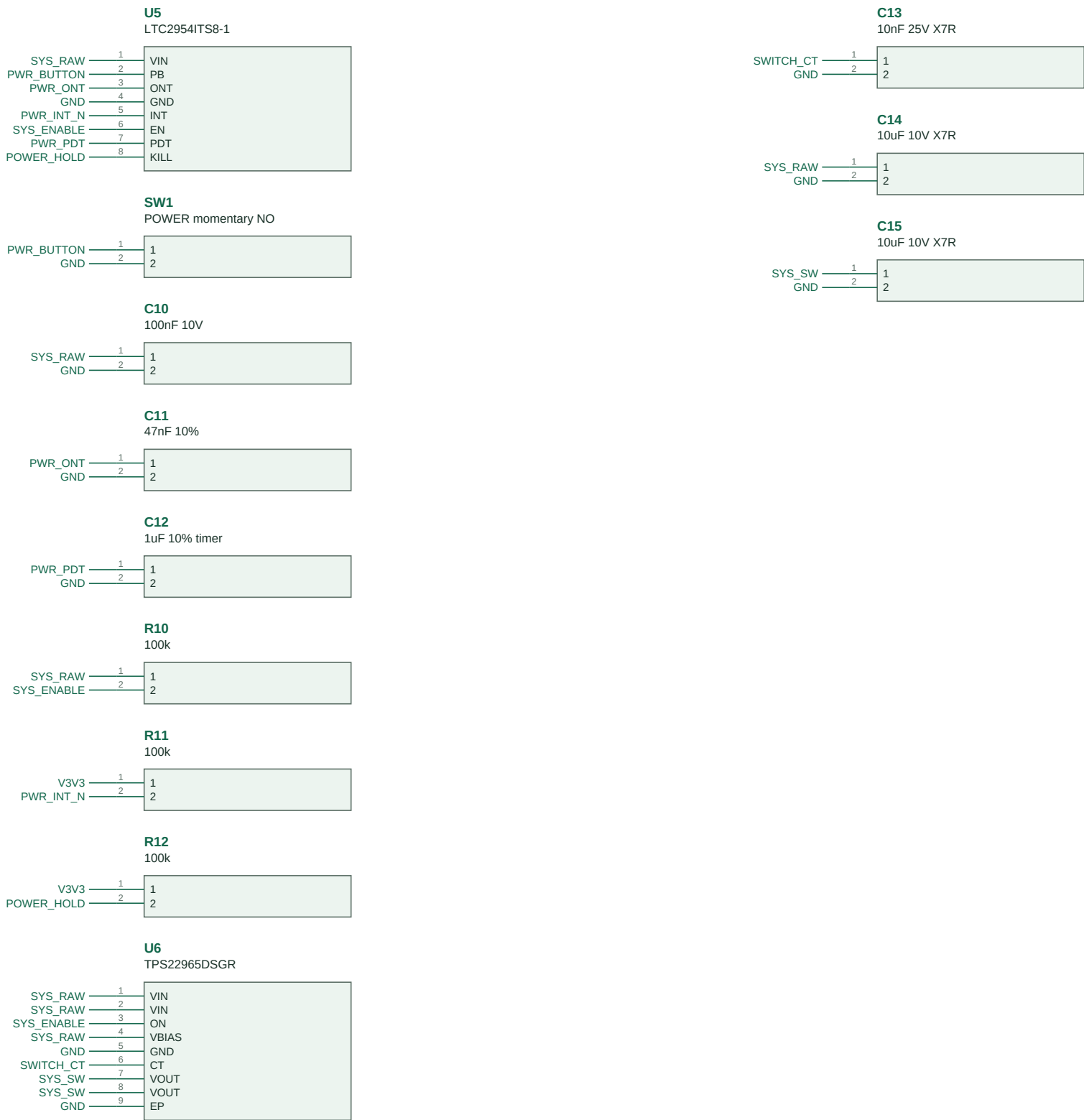
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TUSB320LAI supplies CC termination: do not add parallel 5.1k Rd resistors.
OUT1 low means 1.5 A or 3 A advertised. Inverter selects 1.30 A nominal input limit.
Default advertisement selects USB100. GPIO4 requests USB suspend; implement in firmware.
Battery: protected 1S, 4.2 V charge, >=4 A discharge target, 10k NTC. Existing pack not approved.

Power button and switched system supply

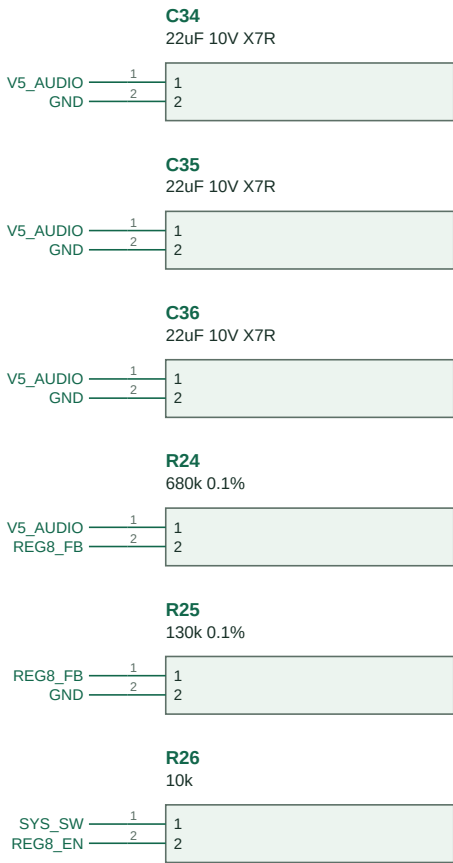
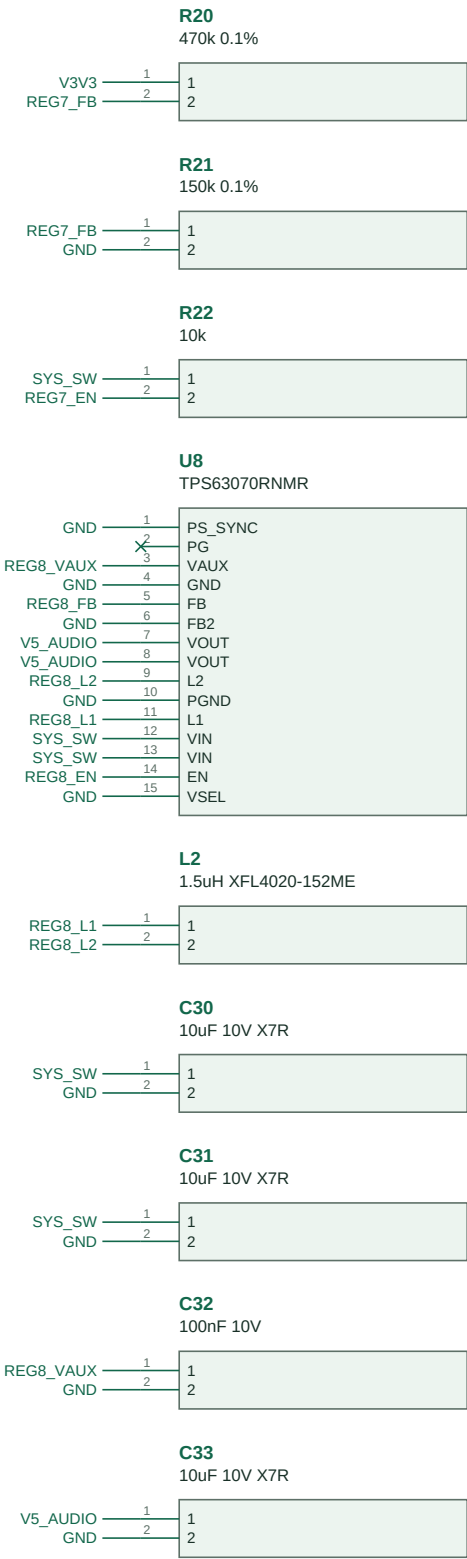
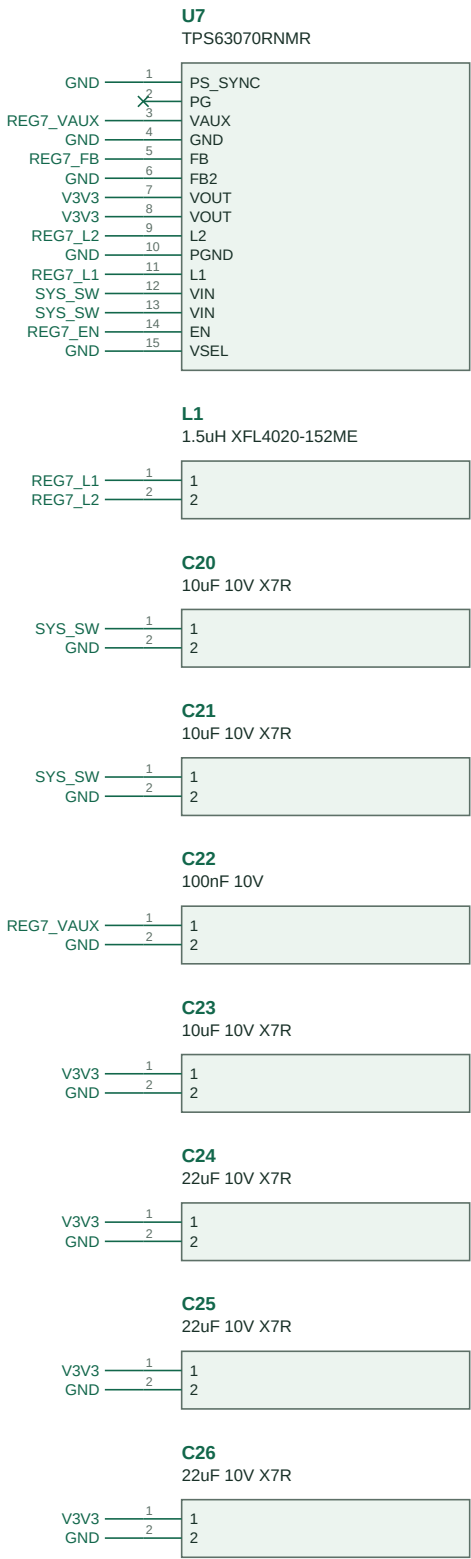
Editable sheet: 02_power_button_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



SW1 is a low-current momentary input; it does not carry battery current.
KILL has a pull-up to switched 3.3 V. GPIO18 must be open-drain; drive low only after SD closure.
Approximate hold times: 0.33 s on; 6.48 s forced off. Normal press raises shutdown interrupt.
Charger/control/gauge remain powered. This is not total battery isolation.

3.3 V and 5 V regulated supplies

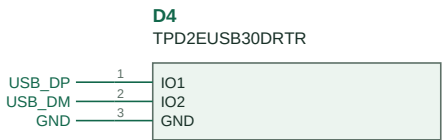
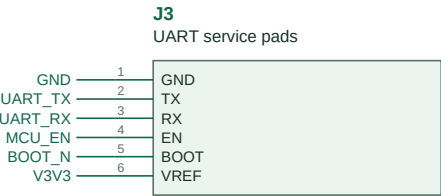
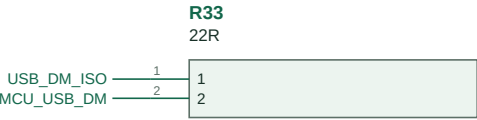
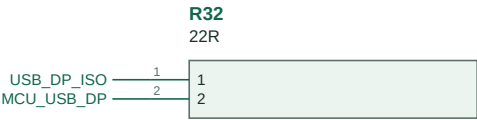
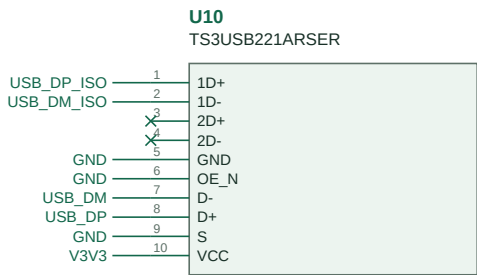
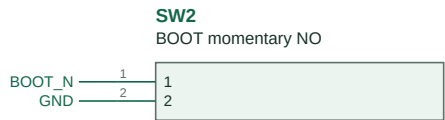
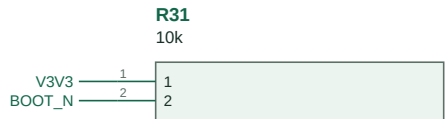
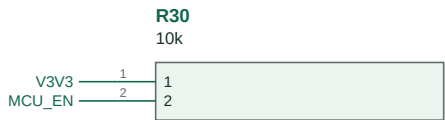
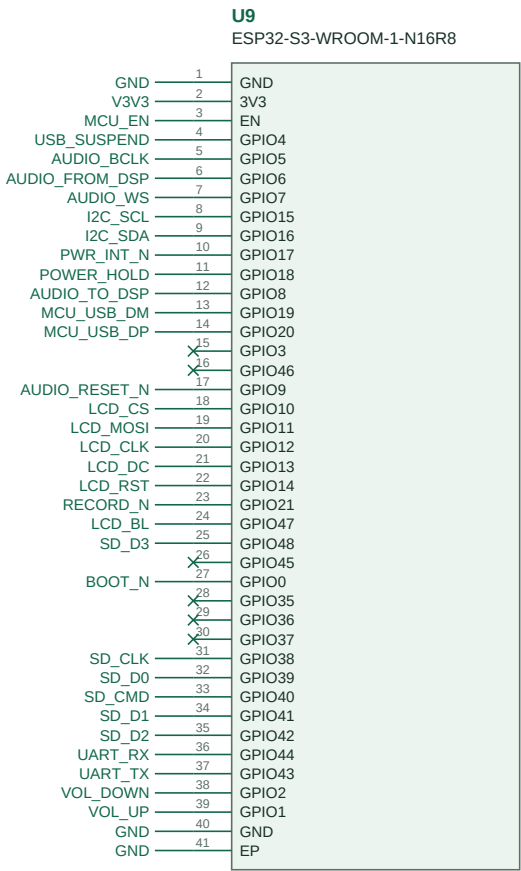
Editable sheet: 03_regulators_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



Both converters use forced PWM, 1.5 uH and 3 x 22uF + 10uF nominal output capacitance.
Verify effective ceramic capacitance under DC bias. Prove loop stability at low battery/full load.
3.3 V: 470k/150k. 5 V: 680k/130k. Values are calculated; tolerance limits are in budget.
L1/L2: XFL4020-152ME candidate. 4.6 A nominal saturation is not an output-current rating.

ESP32-S3, service buttons and USB data

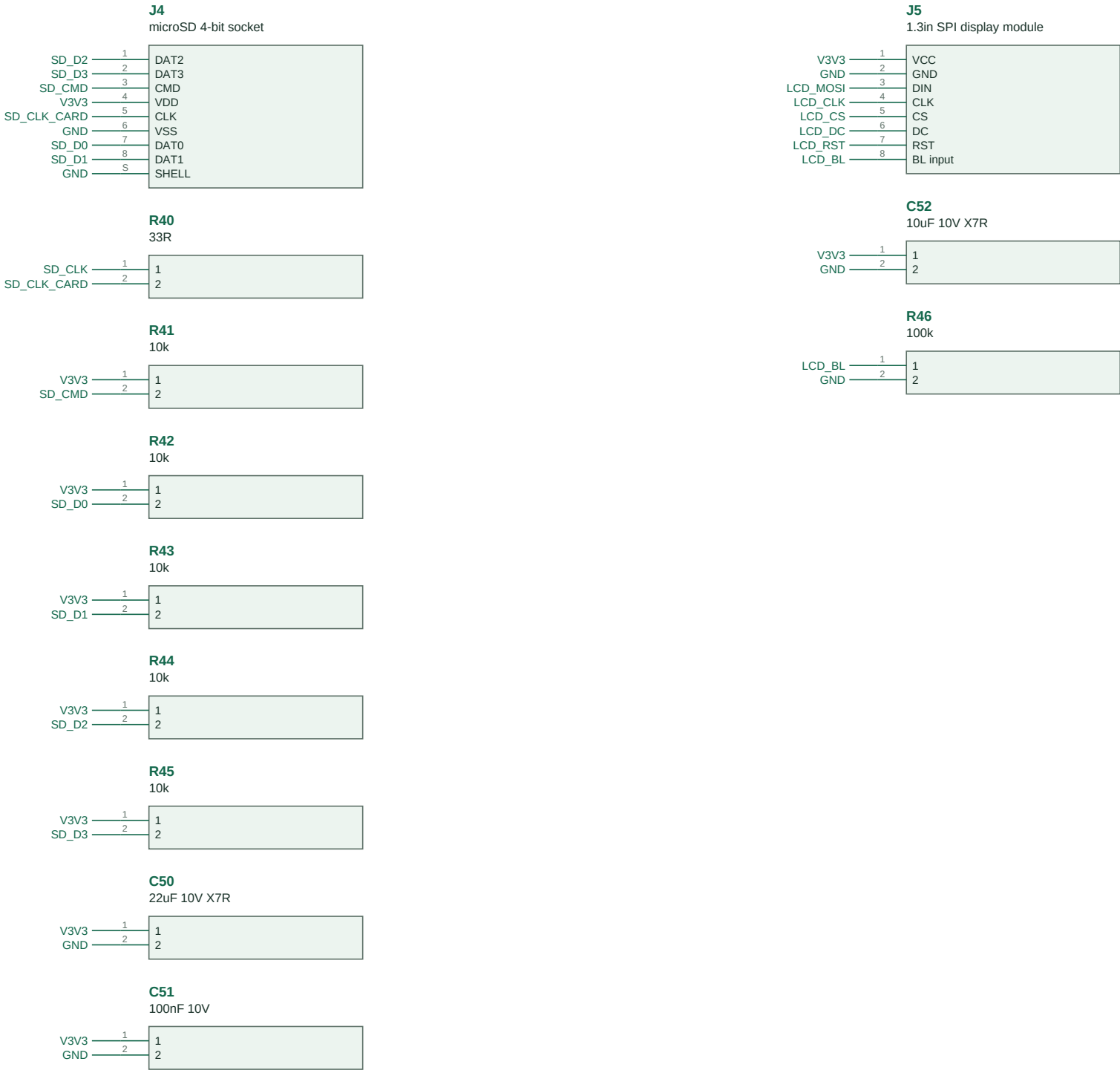
Editable sheet: 04_processor_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



Module pad numbers shown, not GPIO numbers. GPIO35-37 intentionally unused for N16R8.
USB switch provides powered-off isolation; check Ioff and USB eye/ESD after routing.
GPIO4 is USB_SUSPEND, replacing the unused MCLK reserve. No audio MCLK exported.
EN: 10k pull-up / 1uF reset capacitor. Verify reset and converter rise times on the bench.

microSD and small display connector

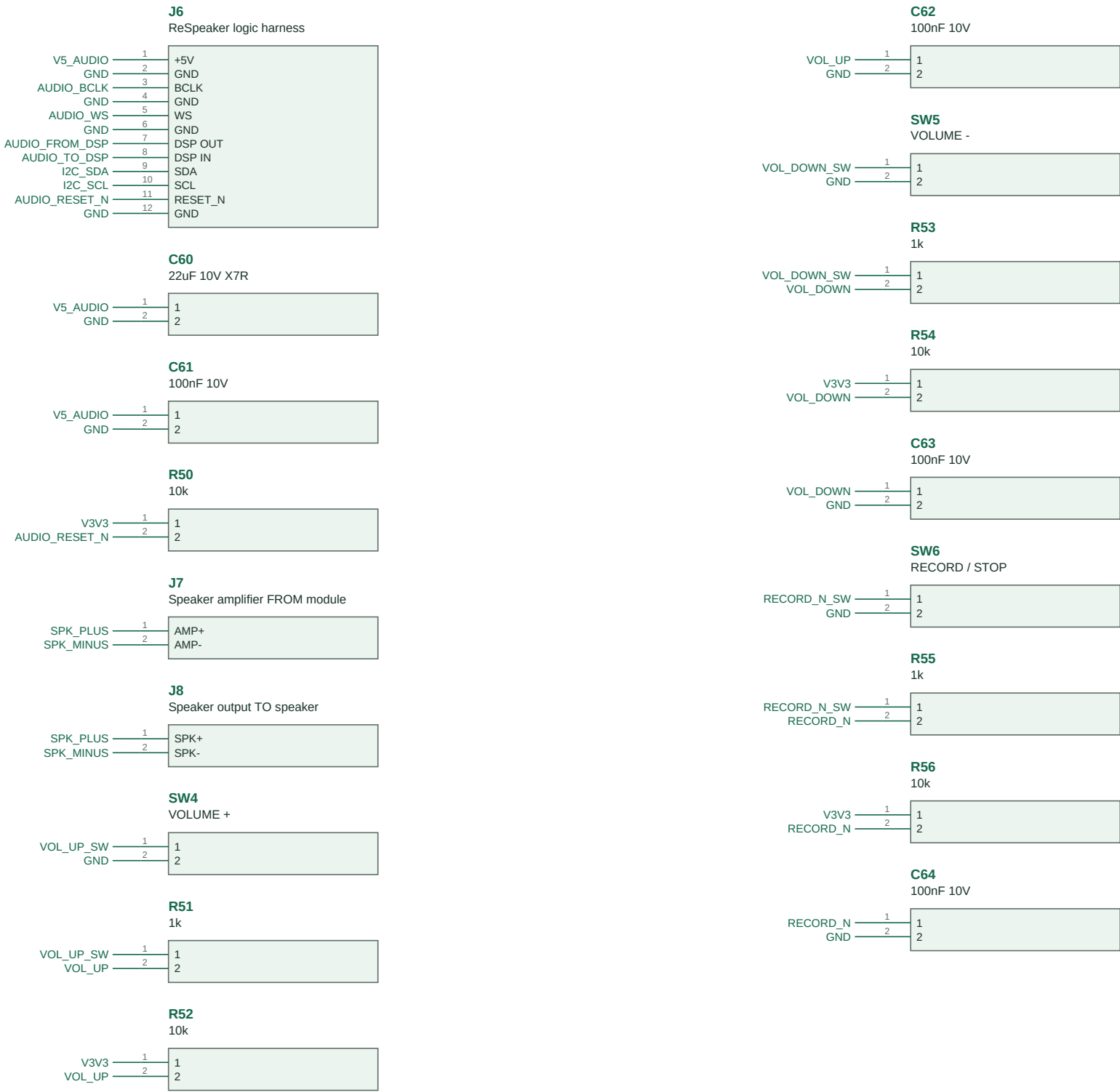
Editable sheet: 05_storage_display_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



J4 numbering is SD card contact numbering; socket detect/shell pad mapping awaits final footprint.
J5 numbering is the carrier connector definition, not a claim about an arbitrary display plug.
Use a 3.3 V-compatible SPI display module with onboard backlight driver; bare glass is not interchangeable.
Display envelope is provisional 42 x 42 x 6 mm. Exact module drawing and mounting holes remain open.

Audio module, speaker and user buttons

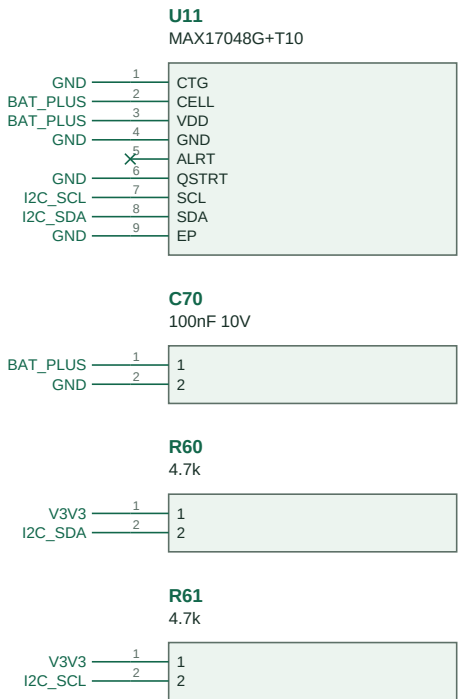
Editable sheet: 06_audio_controls_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



B0 uses four microphones already on ReSpeaker; no four loose external microphones are attached.
J6 is a carrier-defined harness. Map its labels to the selected ReSpeaker revision before assembly.
ESP supplies BCLK/WS for the selected INT-device slave image; verify image clock role before power.
SPK+/- come from the module amplifier. Neither is ground. Target <=2 W into a verified 4-ohm speaker.

Battery gauge and bus pull-ups

Editable sheet: 07_battery_gauge_1.kicad_sch | Matching net labels are electrically connected across all sheets. NC pins carry an X.



MAX17048 uses battery power. I2C pull-ups connect only to switched 3.3 V.
Gauge has open-drain SDA and no battery-rail bus pull-ups. Confirm off-state leakage in assembly.
Address 0x36; check audio addresses. Start at 100 kHz; verify combined pull-up/rise time.
Gauge capacity estimate requires pack characterization; it does not measure audio-board current.

References, files and review handoff

Public manufacturer references checked on 22 September 2026. This document does not redistribute manufacturer schematics.

ESP32 module	https://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf
BQ24074	https://www.ti.com/lit/ds/symlink/bq24074.pdf
TUSB320LAI	https://www.ti.com/lit/ds/symlink/tusb320lai.pdf
Power button	https://www.analog.com/media/en/technical-documentation/data-sheets/2954fb.pdf
System switch	https://www.ti.com/lit/ds/symlink/tps22965.pdf
Both regulators	https://www.ti.com/lit/ds/symlink/tps63070.pdf
USB isolation	https://www.ti.com/lit/ds/symlink/ts3usb221a.pdf
Battery gauge	https://www.analog.com/media/en/technical-documentation/data-sheets/max17048-max17049.pdf
Audio module	https://wiki.seeedstudio.com/respeaker_xvf3800_introduction/
Module mechanics	https://files.seeedstudio.com/wiki/respeaker_xvf3800_usb/respeaker_xvf3800_2d_mechanical_drawing.pdf
I2S example	https://wiki.seeedstudio.com/respeaker_xvf3800_xiao_i2s/
Display candidate	https://www.waveshare.com/wiki/1.3inch_LCD_Module

The package includes STEP and GLB assemblies, a Blender scene, three rendered views, the board-outline STL, editable KiCad schematic sheets, component and pin tables, a calculated power budget, and the generating source. The outline STL is a physical fit dummy only: printing it does not produce an electronic PCB.

Next engineering release: freeze the actual display, battery, speaker and connectors; resolve power stability and thermal results; verify symbols/footprints and native ERC; route and review copper; produce Gerbers, drill, BOM and pick-and-place only after that review.